International Journal of Innovation Studies 8 (2024) 297–312

Contents lists available at ScienceDirect

International Journal of Innovation Studies

journal homepage: www.keaipublishing.com/en/journals/international-journal-of-innovation-studies

Design thinking and artificial intelligence: A systematic literature review exploring synergies

Aswathy Sreenivasan, M. Suresh *

Amrita School of Business, Amrita Vishwa Vidyapeetham, Coimbatore, 641 112, India

ARTICLE INFO

Keywords:
Artificial intelligence
Design thinking
Ethics
Data privacy
Systematic review

ABSTRACT

This article examines how design thinking and artificial intelligence (AI) work together and what it means for the design sector. The goal is to understand how AI technologies may advance the design process, encourage innovation, and produce more individualized and user-centric solutions. This study intends to shed light on the potential of AI as a catalyst for creativity and the ethical implications of AI-driven design by discovering overlapping ideas and methodologies between design thinking and AI. According to the research, AI can significantly influence the design process by eliminating tedious processes, improving user-centricity, and stimulating creativity. AI may support designers' decision-making, prototyping, and ideation processes, resulting in more creative and effective design solutions. Addressing bias in AI algorithms and data privacy is imperative to ensure ethical AI integration. Virtual reality, bio-design, and inclusive design are untapped areas where AI can be used.

1. Introduction

With the rapid progress of technology and societal changes, design thinking has become essential for developing 21st-century skills. It is a commonly used innovation method that affects company culture, customer involvement, and the driving principles of the innovation process. Design thinking was developed as a result of a collaboration between Stanford University and IDEO, with the main goals of laying the groundwork for creative approaches to be in line with corporate goals and helping managers understand designers' thought processes (Liedtka, 2015; Elsbach and Stigliani, 2018; Dennehy et al., 2022; Skov et al., 2022). Artificial intelligence (AI) mimics human cognitive processes. It offers a variety of skills, including perception, logic, learning, problem-solving, and creativity, to complete various challenging tasks (Harika et al., 2022). Because of its adaptability and usefulness across numerous industries, AI boosts output, effectiveness, and problem-solving in diverse contexts (Sharma et al., 2022). The importance of AI in fostering digital transformation cannot be overstated (Brock and von Wangenheim, 2019; Spring et al., 2022; Wamba-Taguimdje et al., 2020). It alters how firms run and promotes innovative methods. As it affects the future of design thinking and the interaction of creativity, it is essential to understand how AI contributes to digital transformation, particularly in recognizing which design thinking stages benefit the most from AI solutions. Design thinking, which encourages innovation by challenging presumptions and coming up with fresh ideas, can work in harmony with AI's ability to process massive amounts of data and identify patterns to provide creative solutions to problems facing the digital revolution.

Design thinking (DT) and AI have become revolutionary in recent years, influencing how we approach problem-solving,

* Corresponding author.
E-mail addresses: aswathyphd@gmail.com (A. Sreenivasan), m_suresh@cb.amrita.edu, drsureshcontact@gmail.com (M. Suresh).

https://doi.org/10.1016/j.ijis.2024.05.001
Received 14 November 2023; Received in revised form 9 February 2024; Accepted 24 April 2024
Available online 24 May 2024
2096-2487/© 2024 China Science Publishing & Media Ltd. Publishing services by Elsevier B.V. on behalf of KeAi Communications Co. Ltd. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

innovation, and technology development (Randhawa et al., 2021). Empathy, creativity, and iterative design processes are prioritized by design thinking's human-centered approach to creating goods and services that successfully meet users' requirements (Mahato et al., 2021). On the other hand, AI has revolutionized many industries by providing intelligent solutions and automating challenging activities because of its capacity to process enormous volumes of data and learn from patterns.

Given that both disciplines have distinctive characteristics that, when joined, could provide potent synergies and unexpected possibilities, the intersection of design thinking and AI offers a potential field for investigation. However, this convergence also brings about difficulties that should be considered carefully. This comprehensive literature analysis intends to shed light on the potential synergies between design thinking and AI while addressing potential integration issues.

Research on the nexus between design thinking and AI is lacking in the literature. Our research attempts to close this gap and support corporate innovation in a highly competitive market. For several reasons, it is essential to comprehend how design thinking and AI work together. First, incorporating data-driven insights can change the development of products and services. It also improves problem-solving by using AI's analytical ability. It additionally customizes AI solutions to improve user experiences. For the benefit of practitioners, educators, and researchers, this study highlights the urgent need to manage the intricate interplay between technological solutions and human-centered design. It allows businesses to use AI and design thinking to make strategic decisions, promoting innovation and competitiveness. Essential ideas, difficulties, recommendations at their junction, and how they promote creativity across fields are key study concerns.

This work examines the relationship between AI and design thinking and its consequences for the design industry. We aim to comprehend how AI technologies may improve design, stimulate creativity, and produce more individualized and user-focused solutions. Through a thorough literature analysis, we aim to identify common ideas and approaches between AI and design thinking. This will help to illuminate AI's potential to stimulate creativity and solve ethical issues in AI-driven design.

We thoroughly examined scholarly databases, concentrating on works addressing the fusion of AI and design thinking. Although our evaluation has limitations, such as potential biases and possible advancements beyond our knowledge cutoff, our analysis offers insightful information about the advantages and difficulties of combining these two fields.

Since there is currently little research on the connection between AI and design thinking, this study fills a gap in the literature. Although both subjects have been thoroughly researched, not much has been said about how they overlap. By carefully examining the body of current research, we want to close this gap and identify areas that warrant additional study.

The ability of design thinking to improve user-centricity, comprehend user wants, and formulate challenges successfully motivates its strategic incorporation into AI systems. This combination speeds development and produces more ethically sound and efficient AI systems by enabling iterative design and fast prototyping. The following sections will detail the practical advantages of this integration, including how it will affect the development of AI, digital transformation, and creative business models.

2. Theoretical background

2.1. Design thinking

Traditional definitions of "design" focus on a product's design and appearance. However, in fields like style, design for interiors, and building design, where products "consist of design," it is becoming more prevalent to connect design to the capacity to resolve complex problems, also known as "wicked problems" (Buchanan, 1992). This idea of design is frequently mentioned in the literature on design thinking. Due to the growing usage of design, there is a need for a system to classify various kinds of design applications.

Design thinking has been debated in the design discourse for the past decade (Hassi and Laakso, 2011; Magistretti et al., 2021; Micheli et al., 2019; Knight et al., 2020), but it was when it entered the management discourse that it attained broad acceptance outside of the design discourse. This is mainly attributable to Tim Brown's efforts (Brown, 2008), who was previously the CEO of the IDEO. Earlier conversations about design thinking were based on academic design research primarily concerned with professional designers' skills and outputs (Buchanan, 1992). According to some, design thinking can help businesses obtain a competitive edge by improving their products and customer experiences or becoming more adaptable and agile (Acklin, 2013).

According to Brown (2008), design thinking is a human-centered approach to innovation that is accessible to people from all backgrounds and is influenced by the ideas and practices of designers. However, it can be challenging to describe because "design thinking" is open to many different interpretations and points of view. This complexity results from various influences, such as conventional design research, current managerial debates, and innovation techniques carried out under the guise of design (Carlgren, 2013). Critical insights into the notion of design thinking are offered by Johansson-Sköldberg et al. (2013), who clearly distinguish "design thinking" from "designerly thinking." While design thinking comprises practices and skills used outside of design contexts by people without formal design training, designer thinking refers to the work and thoughts of professional designers. This clarification adds to Brown's (2008) definition of design thinking and improves understanding. As it examines the practical application of design thinking, their work is especially pertinent in the context of startups and entrepreneurship.

Four principles of cluster design thinking:

(a) Humans as a starting point

Design thinking emphasizes the demands of the individual while drawing ideas and inspiration from people (Schallmo et al., 2018). After identifying these demands, the next stage is to decide whether products and services are technically possible. After that, the cost-effectiveness of the solutions is assessed. The need for significant marketing decreases, and the failure rate of new goods and

services decreases when people and their needs are prioritized.

(b) Multidisciplinary teams

To foster group innovation, design thinking strongly emphasizes the use of multidisciplinary teams with four to six people (Seidel and Fixon, 2013). It is essential to ensure the diversity of knowledge and gender balance. Understanding user perspectives and wants requires empathy. The identification and improvement of solutions are aided by thorough analysis and problem-solving abilities. A great design thinker must embrace innovation, learn from mistakes, and collaborate with industry experts.

(c) Iterative process

Diverse teams with distinctive work processes are ideal for design thinking (Glen et al., 2015). Improving answers requires a defined process that combines logical reasoning with intuition. Divergent and convergent thinking are used throughout the procedure, which involves examining several viewpoints before concentrating on the finest answers. Design thinking is powerful and effective because it produces accurate results through testing prototypes and producing ideas.

(d) Creative environment

The mentioned process occurs in a creative setting, distinguished by a division and institution supporting ideas. Workshop spaces can be altered along with the workplace to encourage creative activity. Resources that can be used in the design thinking process are also pertinent (Lloyd, 2013).

2.2. Artificial intelligence

Today, when a machine exhibits human-like abilities such as reasoning, problem-solving, and learning, we refer to this as AI. However, these skills are used differently than how we learned them (Ameen et al., 2022). Machines may recognize objects they have never seen before, for example, thanks to AI algorithms that have been incorporated into them. The goal of the computer science discipline known as AI is to develop intelligent machines that can carry out tasks that ordinarily call for human intelligence (Simon, 1995). AI has considerably changed over time, spawning a variety of approaches and subfields that have revolutionized a wide range of sectors (de Spiegeleire et al., 2017). In this section, we provide a theoretical basis for some of the most essential AI principles and applications.

2.2.1. Machine learning and deep learning

A crucial component of AI called machine learning (ML) enables systems to learn from experience and improve over time without being explicitly programmed (Carney et al., 2020). It depends on algorithms that understand and analyze data, find patterns, and then make forecasts or judgments based on that analysis. Deep learning (DL), a specialized approach that uses artificial neural networks with numerous layers to process and learn from enormous amounts of data, is one of the most well-known and effective methods in machine learning. Natural language processing (NLP), computer vision, and robotics are just a few fields where deep learning has produced ground-breaking achievements.

2.2.2. Natural language processing

The goal of NLP, a subfield of AI, is to make it possible for computers to comprehend, decipher, and produce human language (Sharma et al., 2020). It entails the creation of models and algorithms to process textual data, enabling operations like sentiment analysis, chatbots, and language translation. NLP applications are increasingly common daily, exemplifying the practical use of NLP technology.

2.2.3. Computer vision

The AI discipline of computer vision aims to enable machines to comprehend and interpret visual data in their environment. Computer vision systems may perform object detection, facial recognition, and image classification using image and video processing methods and pattern recognition algorithms (Brownlee, 2019). These applications have found significant applicability across industries, from driverless vehicles to medical imaging and security systems.

2.2.4. Cognitive computing and decision-making

Cognitive computing aims to use AI techniques to simulate human cognitive processes, including thinking, problem-solving, and decision-making (Wang and Chiew, 2010). These systems use a combination of ML, NLP, and other AI techniques to handle massive volumes of data, analyze intricate situations, and arrive at well-informed choices. Applications for cognitive computing can be found in many fields (Al Mesmari, 2023), including financial risk assessment, strategic business planning, and medical diagnostics.

While combining AI and design thinking results in revolutionary improvements, it poses moral and legal questions. To ensure ethical AI use, privacy, accountability, and regulation are crucial factors (Abrams et al., 2019). Trustworthy AI development and deployment depend on transparency and adherence to industry norms. One must completely understand its ideas and implementations to evaluate AI's interaction with design thinking, revealing its benefits and limitations.

The integration of technology, design thinking, and AI across several fields is the subject of numerous studies. Ebrahimi (2023) encourages technology-assisted, self-directed learning in the classroom. Design thinking is used by Ouyang et al. (2023) to create a therapeutic deep-learning algorithm. Johnson et al. (2023) state that data-driven design will alter design processes. According to Iandoli (2023), it is essential to incorporate design thinking, customer discovery, and lean start-up into entrepreneurship education. Kirongo et al. (2022) combine design thinking and AI to help people with impairments learn and be creative. To provide AI-focused art education in universities, Xu and Jiang (2022) introduced AI-ADT (AI-based Art design and teaching). Using the DiCoT paradigm, Wolferts et al. (2022) investigated how AI might be included in design thinking. The NLG is used by Zhu and Luo (2023) to automate design principles with the GPT. In their 2023 study, Kronqvist and Rousi examined how gender influences AI and the design thinking framework. The potential advantages of AI in industrial design are mentioned by Mahmoud (2022). Siahaan and Tan (2022) investigate variables influencing adaptability during an epidemic in Indonesian businesses. These studies show how technology, design thinking, and AI interact.

2.2.5. Identifying the specific gap in the literature

Despite the expanding literature on AI and design thinking, much ground remains. This study tries to close that gap by examining the interactions between design thinking and AI. These ideas have typically been explored independently, but their combination offers enormous potential since it combines the analytical powers of AI with the user-centered, empathic design approach of design thinking. Our research fills this gap by investigating how combining AI and design thinking can address complex problems in digital transformation and innovation. This research also lays the groundwork for further studies into the dynamic influence of AI and design thinking on innovation, user experience, and ethical issues.

3. Methodology

3.1. Search strategy

We created a structured search strategy based on carefully chosen keywords to perform an extensive and methodical literature study on the connections between AI and design thinking. These keywords were selected because they are commonly used in discussions on related research and are relevant to both domains. We started by conducting an exploratory study of seminal publications in AI and design thinking to identify frequently used terminology and concepts. Through trial searches in specific databases, we gradually narrowed the list of keywords, enabling us to assess the relevance and scope of the results. This procedure aided in the identification of both general and particular terms to include a variety of pertinent studies.

To guarantee thorough coverage of the literature, the search was carried out across the Dimensions database. To efficiently combine terms and create search queries specific to each database's syntax and capabilities, we employed Boolean operators (AND, OR). A typical search string would resemble this, for instance, ("Design Thinking" AND "Artificial Intelligence") OR ("AI" AND "User-Centric Design").

The papers found by these searches were then filtered to find those that specifically discussed the fusion of AI and design thinking using keywords, abstracts, and titles. After that, a full-text examination was conducted to ensure that our chosen articles were pertinent to our study topics. We concentrated on studies addressing the use, difficulties, or results of combining AI with design thinking.

3.2. Inclusion and exclusion criteria

The inclusion and exclusion criteria ensured the selection of pertinent and excellent articles for the systematic literature review. If a paper specifically addressed the intersection or integration of design thinking with AI, it was considered for inclusion. This might include articles examining how AI is used in design processes, how AI is incorporated into design tools, or how AI affects user-centered design. Numerous studies were considered, including conceptual papers, case studies, literature reviews, and empirical investigations. We sought to include a range of viewpoints and thoughts. Due to restrictions on language proficiency, only English-language papers were included. The following exclusion criteria were used:

- Articles not directly related to the intersection of design thinking and AI
- Non-English articles
- Duplicates and redundant publications
- Articles lacking substantial content or scholarly rigor

3.3. Data sources and selection process

After the initial search using the given keywords, 198 documents were found in the Dimensions database. Fourteen documents were removed after applying the inclusion and exclusion criteria because of their scant relevance to the research topic or their duplication across various sources. As a result, 184 documents were ultimately chosen for the systematic literature review.

A thorough study of the research landscape on the interaction between design thinking and AI was conducted by systematically evaluating the literature using the Dimensions database. This analysis highlighted the synergies and obstacles in this developing multidisciplinary subject.

3.4. Methodology for review

We performed a thorough search of the Dimensions database. Systematic reviews are conducted using the Dimensions database since it is more reliable and provides more comprehensive journal coverage compared to Scopus and Web of Science (Achuthan et al., 2022). The validity and dependability of data integrity between Dimensions and other databases have increased through additional studies (Visser et al., 2021). The Dimensions database uses a field categorization technique to obtain reliable field-normalized citation scores (Achuthan et al., 2022). With an emphasis on their intersection, the search phrases employed were taken to find publications that dealt with design thinking and AI. There were many possible papers found in the initial search. We first performed a title and abstract-based screening to weed out papers that did not fit the inclusion criteria. We methodically retrieved critical data for the publications that satisfied the inclusion criteria, including publication details, study goals, methodology employed, significant findings, and contributions to the field. The quality and rigor of each publication were evaluated to guarantee that research of sufficient methodological quality was included in the evaluation. The compiled information was arranged into a logical narrative that highlighted the current status of the subject, significant discoveries, and areas in need of additional study. The discussion and suggestions made in the review are based on this synthesis. This methodical methodology helped us achieve the rigor, thoroughness, and dependability of our literature review, enabling us to make insightful deductions about the relationship between design thinking and AI.

3.5. Limitations

It is critical to recognize some of this study's shortcomings. First, because the study is based on the currently available literature, it might only partially capture some of the intricacies and valuable insights that could be present in actual start-up environments. The study also depends on the caliber and applicability of the chosen publications. Despite the thorough search, some pertinent items should have been noted. Finally, the study excludes primary data-gathering techniques such as interviews and surveys that might offer more thorough insight into the actual experiences and consequences of business owners who employ these tactics.

A protocol must be devised to conduct a systematic literature review because it promotes meticulous "planning," "consistent execution," and "transparency" that enables replication (Sreenivasan et al., 2022). A complete literature review has revealed few protocols (Kumar et al., 2022). Researchers performing systematic literature reviews typically employ the suggested PRISMA or

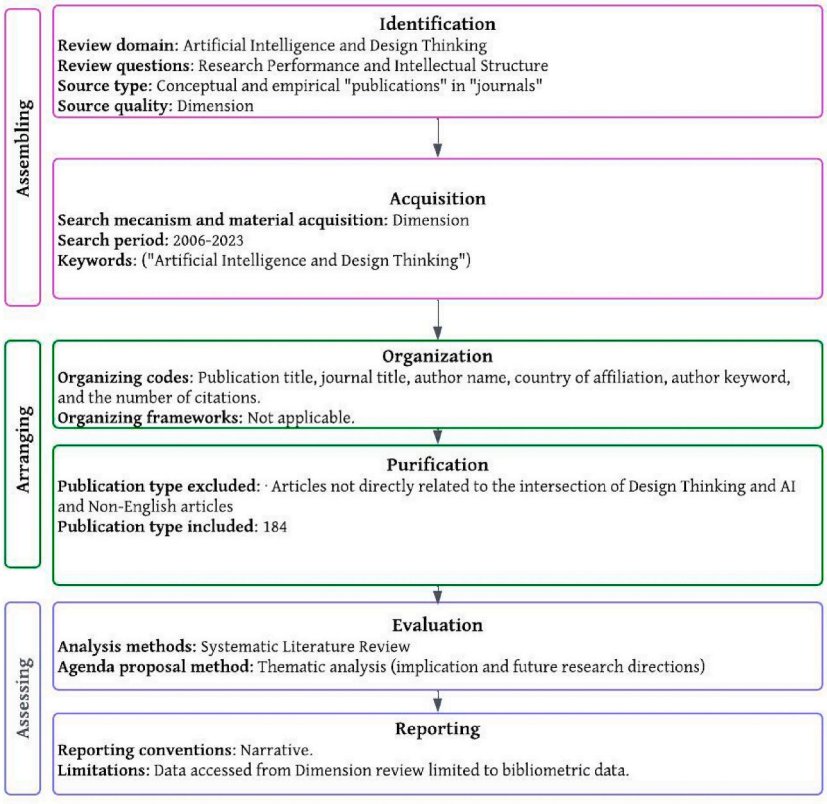


Fig. 1. SPAR-4-SLR.
Source: Paul et al. (2021).

PRISMA-P (Harju, 2022). Although they allow scientists to present their investigations in an ordered, comprehensive, and consistent manner, these two were intended for systematic reviews in general. They provided a few explanations that academics could use to support their study conclusions (Thakur et al., 2022). A protocol called SPAR-4-SLR was developed for systematic literature reviews (Paul et al., 2021). This work used the SPAR-4-SLR's scientific methodology for a systematic literature review (Paul et al., 2021).

3.6. Assembling

Assembling is the first step in the process, which entails gathering publications for analysis. This analysis used the Dimensions database. The search used the keyword mentioned in Fig. 1's title and abstract. After applying all the inclusion and exclusion criteria, 184 publications were included on the shortlist for the years 2006 through 2023.

3.7. Arranging

In the subsequent stage of the process, the publications were removed and sorted utilizing the inclusion and exclusion criteria. The search data for publications was coded using common traits such as "publication title," "journal title," "number of citations," "country of affiliation," and "author name."

3.8. Assessing

The article's evaluation section describes the analytic approach and study constraints. Because secondary data available to anyone with Dimensions access were used, a second ethics clearance was not required. Fig. 1 illustrates the SPAR-4-SLR protocol.

4. Synergies between design thinking and AI

4.1. Identifying overlapping principles and approaches

AI and design thinking are two potent methodologies that have attracted much interest across numerous industries. Each aspect aims to improve the problem-solving and innovation processes, and when used in tandem, they can produce synergies that strengthen each methodology's specific advantages. Let us look at the shared ideas and methods that make design thinking and AI so effective together.

4.1.1. Human-centered approach

Design Thinking: Understanding and empathizing with end users is at the heart of design thinking (de Villiers, 2022). It emphasizes centering the design process around the needs and experiences of people. To develop solutions that genuinely speak to users, design thinkers try to uncover insights into their problems, aspirations, and motivations (Stickdorn et al., 2018). This sympathetic understanding is the basis for developing solutions that connect with users personally. To find unmet needs and insights, practitioners of design thinking employ a range of research techniques, including ethnographic studies, user interviews, and observation sessions. Using these interactions, designers comprehensively comprehend users' circumstances, preferences, and behaviors, which empowers them to devise solutions that effectively address practical issues and augment user experiences.

AI: The caliber and applicability of the training data that algorithms utilize to learn to identify patterns, draw conclusions, and offer recommendations determine how well AI systems perform. Large volumes of user data, including demographics, browsing history, social media interactions, and purchasing activity, are analyzed by AI algorithms. AI systems may customize recommendations, personalize experiences, and anticipate user needs by gaining insights into users' preferences, interests, and behaviors through data mining. Users' data is particularly crucial for machine learning-based AI systems because it allows them to make precise predictions and judgments (Stiglic et al., 2020). The caliber and applicability of the data used to train AI systems frequently determine their effectiveness (Alshamrani, 2022). AI is focused on human-centricity and attempts to offer solutions that complement people's preferences and behavior, much like design thinking.

Integration: Integrating AI with design thinking enhances the human-centered approach by fusing data-driven insights with sympathetic comprehension. By using AI to evaluate user data at scale, designers can find subtle trends and preferences that guide their work. By incorporating AI-driven analytics into design thinking approaches, designers may enhance their decision-making abilities, verify presumptions, and improve solutions by utilizing user input. This collaborative approach guarantees that solutions speak to users' underlying motivations and goals in addition to their explicit demands. By integrating AI's quantitative analysis and Design Thinking's qualitative insights, organizations can develop more user-centric, compassionate solutions that provide significant value to end users.

4.1.2. Iterative prototyping and testing

Design Thinking: The core elements of the design thinking methodology are prototyping and testing (Roberts et al., 2016). Designers produce rapid, low-fidelity prototypes early in development to collect input and validate their concepts. Based on customer feedback, they iterate and improve their solutions, resulting in a more reliable and user-friendly final product. Through iterative testing, designers learn about user preferences, problems, and usability issues. Subsequent iterations are guided by this feedback loop, which helps to match users' better needs through solution refinement. Design Thinking promotes experimentation and iteration by adopting a

fail-fast mindset, which results in more reliable and user-friendly end products.

AI: An analogous iterative strategy has been used in the development of AI, particularly in machine learning (Sarker, 2021). To improve the performance of AI, data scientists have tested different algorithms, model structures, and hyperparameters. AI systems adapt and develop with regular testing and modification, increasing accuracy and productivity (Kelly et al., 2019). AI systems adapt and change through repeated testing and improvement, increasing accuracy, efficiency, and resilience. Iterations afterward are guided by data-driven insights from testing, which help in selecting algorithms and adjusting model parameters. Continued experimentation and modification are necessary to improve AI systems' capabilities and solve performance issues.

Integration: Design thinking and AI are coming together to combine data-driven optimization and human-centered design, which accelerates the iterative prototype and testing process. Using AI, designers can analyze user feedback at scale and conclude massive datasets to guide the iterative design process. AI-powered analytics can help designers prioritize design changes, make better decisions, and shorten the development cycle by being integrated into prototyping and testing workflows. By working together, we can ensure that the solutions are straightforward, easy to use, highly efficient, and effective. Organizations can drive innovation and continuous improvement by combining quantitative data from AI with qualitative feedback from design thinking to modify solutions iteratively.

4.1.3. Embracing ambiguity and exploration

Design Thinking: Design thinkers excel in uncertain circumstances, redefining issues to find original solutions (Uebernickel et al., 2020). They promote creative brainstorming sessions that consider various solutions without instantly limiting choices. The ability to think creatively provides innovative ideas. Ambiguity is ideal for design thinking since it stimulates creativity. This method enables designers to investigate various options without hastily focusing too much. Design thinking makes it easier to find innovative solutions to problems by redefining them outside standard parameters, which may not happen with typical problem-solving techniques. A key component of design thinking is creative brainstorming sessions, which play a critical role in creating an atmosphere that values a range of ideas. Teams can go in various directions during these sessions, which promotes the open exchange of ideas and delays early commitment to a single solution. This willingness to explore ensures that all viable options are considered, which results in more creative and practical solutions.

AI: Large datasets can be handled by AI algorithms, which can also identify trends that people might not see right away. AI algorithms can produce insightful findings and connections that facilitate decision-making and problem-solving by embracing uncertainty and examining various data sources (Jarrahi, 2018). AI handles and makes sense of large, complicated datasets, especially in machine learning and data analytics. AI can find patterns, trends, and correlations in massive amounts of data that human analysts might not notice immediately. Because of this ability, AI can provide fresh perspectives and answers, utilizing ambiguity as a tactical advantage. AI systems can investigate various options from various data sources, including unstructured data (text and images), which is frequently challenging for people to evaluate on a large scale. Through data-driven, non-random research, AI can find new insights to help with problem-solving and decision-making.

Integration: Embracing ambiguity and exploration through merging AI and design thinking offers a potent instrument for innovation. Integrating AI's data-driven research with design thinking's human-centric methodology facilitates a comprehensive investigation of potentialities and obstacles. This synergy makes it possible to use AI's analytical prowess to improve creative brainstorming sessions by offering data-supported insights that might inspire originality and creativity. AI can also help quickly prototype and test concepts produced by design thinking processes, assessing their viability and impact on a large scale. This collaborative approach ensures that the solutions are data-driven and human-centric, successfully solving complex problems in uncertain contexts while speeding up innovation. By integrating AI's analytical and pattern-recognition capabilities with Design Thinking's exploratory and creative qualities, companies may more adeptly traverse ambiguity, resulting in breakthrough ideas and solutions tailored to users' needs and market dynamics.

4.1.4. Data-driven decision-making

Design Thinking: While promoting imagination and empathy, design thinking also emphasizes the value of data-driven insights (Hamington, 2019). Design decisions are informed by user research, feedback, and behavioral data, ensuring that solutions are based on intuition and supported by empirical data.

AI: AI mainly relies on information-driven judgment. Applications for AI must have access to pertinent, high-quality data to succeed (Krishna et al., 2023). Organizations can get a deeper understanding of their consumers and develop more individualized and efficient AI-powered solutions by combining the method of user data collection used by design thinking with AI's data analytics capabilities. By combining AI's data analytics capabilities with the user data collection techniques used in design thinking, organizations may use AI to better understand their customers. AI may rapidly and effectively process and analyze large datasets, yielding actionable insights that can lead to effective and customized solutions. Organizations may make data-driven decisions at scale and with more significant effects by incorporating AI into the decision-making process.

Integration: Both the qualitative and quantitative parts of decision-making processes are improved when design thinking and AI are combined with data-driven decision-making. Design thinking offers a human-centered approach, guaranteeing that solutions are intuitive and sympathetic, while AI offers the analytical power to scan and evaluate large datasets quickly. Organizations may create more intelligent and practical solutions based on user demands and preferences and empirical evidence by combining the best features of both techniques. With the help of this cooperative approach, businesses can make data-driven decisions customized for each user, improving user experiences and boosting productivity and commercial results.

4.1.5. Scalability and automation

Design Thinking: Implementing design solutions has typically been labor-intensive and manual. However, the scalability of design thinking can be significantly improved by introducing AI automation (Verganti et al., 2020). Within the design thinking process, AI may automate repetitive and time-consuming processes such as user data analysis, feedback pattern identification, and even the creation of early prototype ideas. By automating these processes, designers may concentrate more on creative exploration and nuanced problem-solving, where human understanding and empathy bring invaluable value. AI can also replicate user interactions with prototypes, resulting in broader and faster feedback loops that improve iterative design.

AI: Automation of tedious work and rapid data processing are two areas where AI thrives. AI can be used to speed processes like data analysis, pattern recognition, and even the generation of design recommendations, allowing designers to concentrate more on the creative aspects of their work (Schneider et al., 2020). Moreover, AI's automation power extends to large-scale customization of design solutions. AI systems can analyze user data to generate individualized user experiences by customizing features, interfaces, and content to each user's preferences. With this degree of AI-powered personalization, Design Thinking can reach a wider audience and make a more significant difference by catering to its unique needs and preferences.

Integration: Design Thinking's human-centered approach and AI's automation capabilities work together to provide a potent platform for innovation. AI's automation and scalability facilitate quick iteration and modification of design solutions, making it possible to apply design thinking more successfully and efficiently to various issues and situations. By utilizing AI, the design thinking process becomes more flexible and dynamic and able to address large-scale, complicated problems while remaining focused on providing solutions that are important to the user. In addition to improving the efficacy and efficiency of the design process, this integration allows firms to innovate and react faster to shifting consumer demands and market conditions. To summarize, AI and design thinking work together to improve scalability and automation, which opens the door to more creative, customized, and responsive design solutions. This dramatically broadens the use and influence of human-centered design techniques.

The fusion of design thinking and AI combines human-centricity, iterative problem-solving, and data-driven insights. This combination can produce more creative and practical solutions that meet user needs and use AI to process and analyze massive amounts of data. Organizations can maximize the benefits of this dynamic synergy by adopting these disciplines' shared values and methods.

4.2. Leveraging AI to augment the design process

AI has the potential to improve design significantly by providing designers with additional resources, perspectives, and talent to produce more original and practical solutions. Designers may expedite repetitive activities, improve decision-making, and obtain deeper insights into customer demands by incorporating AI technology into various phases of the design workflow. AI can be used to improve the design process in several ways.

4.2.1. Ideation and inspiration

AI algorithms can help designers explore various design options by developing other solutions based on particular requirements or preferences (Rosenschein and Zlotkin, 1994). This may promote creative and surprising design directions. AI may help designers by offering a vast database of design references, styles, and trends during brainstorming (Brown et al., 2020). Large databases of existing designs can be analyzed by AI-powered tools, which can then produce concepts based on predetermined criteria, giving designers ideas and a place to start.

4.2.2. Rapid prototyping

AI can speed up the prototyping process by automating the production of low-fidelity prototypes based on design parameters (Brown et al., 2020). Faster development cycles are made possible by this, allowing for quick iterations and testing of design concepts.

4.2.3. Design automation

AI can automate monotonous design activities, such as creating different layouts, color schemes, or typeface options (Trocin et al., 2023). Due to this automation, designers can now concentrate on more imaginative and strategic aspects of the design, saving them time.

4.2.4. Data-driven design decisions

User data, market trends, and performance indicators can be processed using AI-driven analytics to help design decisions (Alghamdi and Al-Baity, 2022). Using these insights, designers can develop solutions that align with user preferences and improve designs for improved user experiences.

4.2.5. Collaborative design

AI can make collaboration easier for design teams by offering a standard hub for exchanging concepts, suggestions, and design files (Marshall, 2009). Regardless of team members' location, AI-driven collaboration solutions can enhance communication and coordination.

4.2.6. Design quality assurance

AI can be used to automate quality checks of designs to ensure that they follow set standards and norms (Rosenschein and Zlotkin, 1994). This keeps the brand's identity and consistency across all design materials.

By using AI to support the design process, designers may become more effective, imaginative, and data-driven. By using AI technology, designers are free to concentrate on higher-value jobs, while AI takes care of the analytical and repetitive parts of the design. Ultimately, more reliable, user-centric, and creative design solutions are produced due to the interaction between human ingenuity and the computing ability of AI. Designers must remember that AI is merely a tool and that human touch is still necessary to infuse the design process with empathy, emotion, and ethical considerations.

4.3. AI-driven personalization and customization in design thinking

Using user data to learn preferences and pain spots, AI-powered personalization and customization improve design thinking. This data-driven methodology develops thorough user personas, which enhances comprehension of consumer needs. Designers can use AI recommendation algorithms and dynamic content production to create tailored user experiences that respond to context and individual preferences. By providing pertinent content and recommendations, this level of personalization improves user-design interactions (Yuan et al., 2023; Haleem et al., 2022).

Through AI predictive algorithms, designers can foresee customer demands and possible problems, proactively address them with design solutions before they materialize, and ultimately increase user pleasure and loyalty (Jarrahi, 2018). Iterative AI-driven personalization enables continual improvement by examining user comments and behavior. AI facilitates active user participation, encouraging cooperation and insightful decisions (Solakis et al., 2022). Additionally, AI's emotional design analysis provides users with insights into how design aspects affect their emotions, enabling the development of emotionally engaging experiences.

Design thinking has much potential for incorporating AI-driven personalization, but carefully handling user data and privacy is crucial. To provide comfortable, personalized experiences, trust-building and securing user consent is essential. In conclusion, combining AI-driven personalization with design thinking presents the chance to develop products considering customer preferences and wants. This partnership eventually increases client happiness and corporate success by enhancing design experiences and making them more potent, captivating, and user-centered.

5. Ethical and societal implications

5.1. Addressing bias and fairness concerns in AI-driven design

AI has many advantages but also raises ethical questions about prejudice and fairness, as it is integrated more deeply into design. The AI-driven design has the potential to be biased, which is a severe concern because it could reinforce biases found in training data. The use of fairness-aware algorithms, diverse and representative training data, and data preprocessing are all stressed by designers as ways to reduce prejudice. AI model transparency is essential for detecting and comprehending biases. AI models must be regularly monitored and audited to identify and fix problems early. The importance of inclusion in the whole design thinking process is highlighted by inclusive design principles, which ensure that various user demands are considered.

Although AI offers valid data, human designers must ultimately make decisions (Yeung et al., 2020). AI does not possess humans' context or empathy, guaranteeing that the design process is moral and considers all possible outcomes. To identify biases that may be concealed, it is essential to encourage user feedback. Organizations should be open to using AI and efforts to reduce bias. Designers must abide by moral standards and data privacy regulations. Responsible AI use is encouraged by including ethics in AI design techniques (Floridi et al., 2021). Identifying potential prejudice and blind spots is made more accessible through collaboration with ethics, sociology, and human rights specialists. Responsible AI-driven design requires staying current with ethics and fairness studies. Moral and trust-building necessity, prejudice, and justice in AI design must be addressed. Designers must use AI ethically and creatively to provide inclusive and fair designs that reflect social ideals.

5.2. Impact of AI on employment in the design industry

Using AI in the design sector can positively and disruptively impact job dynamics. Although AI-based technologies can increase output, improve procedures, and open up new possibilities, they may also result in certain job losses and changes to the duties of design experts. Data processing, layout development, and color choosing are labor-intensive design processes in which AI automation is optimized, allowing designers to concentrate on creativity and strategy. By providing data-driven insights and novel ideas, AI tools improve skills and encourage collaboration between designers and AI for more creative designs (Han et al., 2020). New positions, including those of AI professionals, data analysts, and AI ethics experts, are also created due to the integration of AI in design (Rezwana and Maher, 2022). To be competitive as AI takes over specific duties, designers may need to modify their skill sets, including data analysis and ethical issues. Routine, rule-based design tasks are more amenable to automation by AI.

To ensure ethical and unbiased AI use, AI raises ethical problems and changes the roles of designers. AI and designer collaboration change the typical client-designer dynamic. This could impact the demand for design services and even open doors for AI experts to join design teams. To remain relevant in the changing AI-driven design market, designers must continually learn new skills and improve their knowledge. Finally, AI will have an impact on a variety of design tasks. Some roles may be replaced, but they also provide designers with cutting-edge tools and abilities, opening up new prospects. The degree to which designers adopt and advance their AI skills will influence how well AI is incorporated into the design sector. As with any technology change, embracing AI's potential benefits while limiting job disruptions requires a proactive and cautious strategy.

5.3. Privacy and security issues in AI-infused design thinking

Significant privacy and security issues are brought up by integrating AI technology into the design thinking process, which must be appropriately addressed. Data security and privacy protection are crucial since AI-driven design relies on a tremendous quantity of data, including user data and behavioral patterns.

Designers must emphasize data privacy in the context of AI and design thinking, abide by pertinent data regulations, apply robust security measures, address biases, maintain transparency with users about data collection and usage, and limit data collection to what is necessary for the design process. Designers must be cautious about the privacy policies of third-party AI tools and service providers if they employ their products. To protect user data, it is essential to ensure that third parties follow the same privacy laws. Obtaining the user's explicit agreement is essential when employing AI-driven design solutions. Users should have the choice to decrease data gathering or personalization based on AI if they want. Data sharing may be a feature of collaboration between design teams and outside partners. Secure data-sharing methods and anonymization procedures should be used to safeguard user privacy. AI systems are vulnerable to adversarial attacks, in which malevolent parties knowingly modify data to influence the system's decisions. Designers must know these weaknesses and implement defenses to thwart such assaults. After implementation, the privacy and security of AI-driven design solutions should be continuously monitored and assessed. Regular security audits and vulnerability assessments are necessary to identify and address potential threats.

Designers should take a privacy-by-design approach, prioritize ethical concerns, and collaborate closely with data security professionals to reduce privacy and security issues in AI-infused design thinking. All parties participating in the AI-driven design process must continue their education and understanding of privacy and security best practices. By proactively addressing these challenges, designers may develop novel, user-centered solutions while maintaining users' privacy and data security.

6. The significance of integration: Bridging the gap in digital transformation and creative business practices

Integrating design thinking with AI offers a fundamental paradigm change with far-reaching ramifications in today's fast-changing digital transformation landscape and innovative business practices. Despite the well-established significance of each subject separately, their synergistic integration provides distinctive solutions to current problems and closes significant gaps in current approaches. Empathy, user-centric design, and creative problem-solving are prioritized in the human-centric approach of design thinking. On the other hand, AI offers automation and data-driven insights. The merging of these two domains fills the gap between human creativity and data-driven decision-making. It enables companies and organizations to produce goods and services that are not only cutting-edge but also acutely aware of the requirements and preferences of users, ultimately improving user experiences. Innovative business practices frequently rely on ideas, experience, and intuition. The ability of AI to handle enormous volumes of data and identify patterns opens up new possibilities for data-driven innovation. By fusing AI and design thinking, decision-making may be made more intelligently and creatively by equipping creative professionals with tools for data analysis and predictive modeling. Processes that are simplified and scalable are frequently required for digital transformation. Automating tedious processes and streamlining workflows are areas where AI thrives. This integration increases the effectiveness of creative procedures and guarantees that scalability does not compromise user happiness when combined with the iterative, user-focused techniques of design thinking. Initiatives for digital transformation usually run into complicated, diverse issues. While design thinking offers a framework for original problem solving, AI can examine these issues. Together, they provide a comprehensive strategy for resolving problems that neither strategy alone can solve. Iterative methods that frequently involve prototyping, testing, and refining are often what creative businesses need to succeed. The iterative nature of design thinking fits nicely with AI's capacity for continuous learning and adaptation. These two approaches work in harmony to promote dynamic and flexible innovation processes. We bridge the gap between human creativity and data-driven insights and address the urgent issues in digital transformation and innovative business practices by combining design thinking concepts with AI. The ability to produce user-centric, morally sound, and innovative solutions that fit the intricate requirements of the contemporary business landscape is provided by this confluence. In the following sections, we will go into more detail about how this integration is used and its effects on different areas, offering insights into its practical significance.

7. Examples of start-ups successfully incorporating AI and design thinking

Successful startups that have combined design thinking and AI include the following:

7.1. PillPack

The drug dispensing process is personalized and automated by PillPack, a pharmacy company provides its clients with a user-friendly experience; PillPack also applies design thinking. PillPack, a new pharmacy, aims to modernize this antiquated procedure. PillPack will deliver prescription medications to customers at Amazon Prime speed for $20 per month. The foundation of Pillpack's service is a small blue box that organizes all of your medications into "dose packets," small plastic baggies labeled with the date and time they are to be taken. Pillpack was born from a new incubation program at the renowned design consultant IDEO. Instead of a tangle of amber bottles, a helpful to-do list constructed of medications is substituted.

7.2. Airbnb

Airbnb is a new firm that rents homes and utilizes AI to pair visitors with the ideal lodging. To provide users with a smooth experience, Airbnb also applies design thinking. A website experience such as Airbnb's is unmatched in the travel industry. By changing how people travel, Airbnb's user experience has become smoother and more satisfying. The brand spends much money on user research to ensure that the website is user-friendly, easy to browse, and meets the demands of its visitors. More importantly, Airbnb was carefully created based on consumers' trust. They tried to transform the customer's anxiety about spending the night in a stranger's home into helpful information. By providing precisely the right amount of information, including facts about the home, the host, the neighborhood and location, and its proximity to major landmarks and modes of public transportation, Airbnb created a foundation of trust. The brand ensured hosts had a long list of amenities and comforting information to share with guests. Their host list form was created with an intuitive link to boost host acceptance.

7.3. Spotify

A music streaming startup, Spotify, utilizes AI to make song recommendations to its subscribers based on their listening preferences. To provide users with a tailored experience, Spotify also applies design thinking. For instance, Spotify employs design thinking to create playlists customized to each user's preferences. The sleek, contemporary design of the Spotify app and website complements the look and feel of the business. With a straightforward, basic design, Spotify's UX design promotes simplicity and ease of use, making it simple for consumers to find and access the desired music. Users' engagement is boosted, and cognitive burden is reduced. Users enjoy a distinctive and customized experience because of Spotify's personalization, making it simple to find new music they might not have otherwise come across. This increases user engagement and pleasure. The usage of micro-interactions is a critical component of Spotify's UX design. Micro-interactions, such as a button click or a swipe gesture, are brief, undetectable animations or interactions that offer feedback or acknowledge a user's activities. These small-scale interactions make the experience more engaging and fulfilling, providing the user with control and feedback.

8. Enhancing design creativity through design thinking and AI

AI has been a game-changing technology in the design business in recent years, significantly impacting and increasing design creativity in various ways. The relationship between design thinking and AI is examined in this section, emphasizing how their combination encourages innovation and creativity in design processes.

8.1. Design communication and visualization

Throughout the design process, design thinking strongly emphasizes effective communication and visualization (Carlgren et al., 2016). The depiction of design concepts is made more accessible by AI technologies like generative design algorithms and augmented reality (AR) tools, which help designers communicate ideas more clearly and persuasively (Vilar et al., 2022). Designers can explore and express complex design concepts more successfully by utilizing AI-powered visualization tools, encouraging innovation and cooperation across interdisciplinary teams.

8.2. Facilitating visual thinking

A key design process component is visual thinking, which allows designers to generate ideas, refine them through iterations, and present them visually (Bresciani, 2015). By automating processes like pattern identification, visual synthesis, and image analysis, AI-powered solutions like computer vision algorithms and image recognition systems enhance visual thinking (Sarker, 2022). Using AI to enhance visual thinking processes, designers can uncover novel insights, unexplained patterns, and inventive design alternatives, ultimately improving their creativity and ideation (Liao et al., 2020).

8.3. Generating stimuli and inspiration

AI algorithms can examine enormous volumes of design data, such as pictures, sketches, and historical design patterns, to provide designers with ideas and inspiration (Liu et al., 2020). AI-powered recommendation systems can stimulate designers' creativity by spotting patterns, trends, and up-and-coming looks. This can lead to fresh concepts and inventive avenues. Furthermore, AI-driven tools for design generation, such as generative adversarial networks (GANs) and creative AI algorithms, can produce design concepts, prototypes, and variations on their own (As et al., 2018). These tools can be an excellent way for designers to explore and become inspired.

8.4. Managing knowledge and insights

To inform design decisions, design thinking depends on synthesizing and managing varied knowledge and insights (Carlgren et al., 2016). Semantic knowledge graphs and intelligent search engines are two examples of AI-powered knowledge management systems that help designers access, organize, and synthesize design-related data. By using AI to manage knowledge and insights, designers may

make better judgments, draw from a broader range of sources, and develop a deeper understanding of design principles and processes. This increases creativity and innovation.

8.5. Generating content and design assets

AI technology, like automated design systems and NLP algorithms, may create content and design assets on their own (Carvalho et al., 2019). Based on predetermined standards and constraints, these AI-driven content-generating technologies can generate written content, graphic designs, and multimedia assets (Quan et al., 2023). AI helps designers concentrate their creative energy on higher-order tasks, such as concept generation, ideation, and design strategies, by automating repetitive chores and producing design materials at scale. This increases overall design creativity and productivity.

There is much promise for improving design creativity and innovation when AI and design thinking are combined. Designers may uncover new creative possibilities and push the frontiers of design excellence by utilizing AI technology to assist in design communication, facilitate visual thinking, produce stimuli and inspiration, manage information, and automate content development. More research and discovery in this multidisciplinary field are necessary to thoroughly understand the revolutionary impact of AI on the design business.

9. Comparative analysis with existing approaches

Comparing our suggested synergies between design thinking and AI with current methodologies in this area is a crucial component of our research that needs more work. This comparison is crucial to defining our framework's distinct contributions and placing it in today's context of the state of knowledge and practice.

9.1. Traditional design vs AI-enhanced design thinking

Conventional design approaches rely significantly on human creativity, intuition, and iterative prototyping and testing cycles. These techniques are efficient but can take much time and do not always use data-driven insights. Our method combines AI with conventional design thinking procedures, enabling faster testing, iteration, and prototyping based on data-driven insights. Because of this synergy, design processes are more effective and can respond more quickly to user input and new trends.

9.2. AI in isolation vs AI in design thinking

Traditionally, AI applications in design have concentrated on automating particular processes, including producing design alternatives or optimizing design parameters. Despite their strength, these apps frequently function independently of the human-centered elements of the design process. We suggest a more comprehensive strategy in which AI technologies are directly integrated into the identify, empathize, ideate, prototype, and test stages by integrating AI within the design thinking framework. The total originality and efficacy of design solutions are increased by this integration, which guarantees that AI-driven insights and capabilities are consistently aligned with human-centered design principles.

9.3. Standalone design thinking vs integrated DT-AI framework

As a stand-alone paradigm, design thinking emphasizes human-centric problem-solving, which may not fully use the potential of cutting-edge technology like AI, even though it is effective at promoting innovation. Our combined DT-AI framework combines AI's efficiency and predictive capability with the human-centric approach of design thinking. This leads to the creation of an inventive and cutting-edge problem-solving toolset that is more extensive.

9.4. Comparison with specific baselines

Examine the application of AI in user experience (UX) design without the DT framework for a concrete analogy. While these methods can employ user data and prediction models to guide design choices, they do not follow a formalized procedure for iterative learning and empathy-driven design. Our methodology incorporates AI for data analysis, prediction, and the iterative cycles of empathy, ideation, and prototyping inherent to design thinking. This guarantees that technological innovations are constantly based on a thorough comprehension of the requirements and environments of their users.

10. Resource limitations

The main constraints of this study are its limited resources, which include time, money, and data access. The available data collection, processing, and interpretation resources may have limited the breadth of the evaluation. Consequently, it is possible that specific pertinent research or viewpoints were excluded from the evaluation, which could have limited how thorough the results were. Furthermore, the breadth of the analysis or the capacity to carry out empirical research to support the theoretical insights offered in the study may have been limited by a lack of resources.

Another critical constraint is the dynamic and quickly changing nature of AI technology and ethics. New algorithms, approaches,

and applications are being developed quickly in AI, which could make some of the conclusions and insights in the study outdated or lacking. Furthermore, as cultural norms, laws, and ethical frameworks change, so do the ethical concerns surrounding AI, such as prejudice, justice, privacy, and responsibility. As a result, when research into AI and ethics advances, the interpretations and ramifications of conclusions might also change.

These limitations significantly impact the interpretability and applicability of the findings of this research. Researchers and practitioners need to recognize how resource limitations may have affected the breadth and depth of the literature. They should also be aware of how AI ethics and technology are evolving and be careful to keep their understanding and practices up to date. By being aware of these constraints, researchers may ensure that their conclusions are solid and useable in various contexts and offer a more nuanced and contextualized interpretation of their findings.

To summarize, a more thorough examination of the limitations—such as those related to resources, the dynamic nature of AI technology, and ethics—is necessary to put the research's findings into context and increase its legitimacy and applicability. Researchers may help develop a more sophisticated and nuanced understanding of how AI and design thinking intersect to foster design innovation by openly discussing these limitations.

11. Future research direction

11.1. Emerging trends and opportunities

AI and design create new trends and opportunities (Lee and Park, 2019). AI-augmented creativity investigates how AI might improve concept generation and design optimization, providing new ideas for human designers (Zhou et al., 2023). Explainable AI in design strives to increase the transparency and understandability of AI-driven decisions, promoting cooperation. Design experiences that are more emotionally engaging are produced through AI-driven emotional design, which focuses on understanding and responding to human emotions (Huang and Rust, 2021). Data from multiple sources are combined with multi-modal AI in design, improving immersive design solutions. Finally, AI for Sustainable Design explores how AI might support eco-friendly and energy-efficient goods, encouraging eco-aware design principles.

11.2. Unexplored domains and application

The advancement of AI has created new opportunities for the design sector. AI can improve spatial design, object detection, and interactive interfaces for immersive experiences in virtual and augmented reality design (Cao et al., 2023). The utilization of materials is optimized, functionality is improved, and creative ideas are produced using AI-driven generative design (Badini et al., 2023). AI-enhanced bio-design blends AI and biotechnology for applications involving biomimicry and bio-inspired solutions. Accessibility for a range of users is improved through AI for inclusive design (Worsley and Bar-El, 2022). AI-driven sound and music design transforms soundscapes through interactive audio experiences and AI-generated musical compositions.

11.3. Interdisciplinary perspectives and collaboration

Interdisciplinary cooperation is essential to the success of AI-driven design (Shi et al., 2023). While collaborating with experts in AI and neurology, cognitive scientists offer insights into designing human-centered AI systems and investigating the brain's function in creativity (Dang et al., 2023). For ethical frameworks in AI-driven design, ethicsists are essential (Bleher and Braun, 2023). AI and environmental science can be used to address sustainability. Collaboration with experts in healthcare design improves patient-centered settings and medical technology (Reay et al., 2017). Adopting interdisciplinary viewpoints guarantees that AI is used ethically and creatively, which benefits many industries and society (Chassang et al., 2021).

12. Conclusion

This study examined how AI and design thinking work together to improve design by making it more unique and innovative. It made clear how crucial it is to address issues of justice, bias, and ethics when incorporating AI into design processes. The study also revealed potential future fields of study and cross-disciplinary collaborations influencing AI-infused design thinking. The design business has much to gain from combining design thinking and AI. Responsible AI-driven design results from embracing human-AI collaboration focusing on ethics and user privacy. This strategy stimulates creativity, improves user experiences, and benefits society. By proactively and morally incorporating AI, design thinking can transform the design process, improve lives, and contribute to a more inclusive and sustainable future.

The originality of this study resides in its thorough analysis of the use of design thinking and AI, including new developments, moral considerations, and uncharted territory. This study emphasizes the potential of AI as a catalyst for innovation, creativity, and user-centric design while simultaneously stressing the significance of moral and proper AI practices. This study advances the knowledge and use of AI in design by offering insights into future research areas and interdisciplinary collaborations, paving the way for a more creative and inclusive future.

While this study investigates the potential of AI-infused design thinking, it is important to recognize several restrictions that limit how our results may be interpreted and applied more broadly. These restrictions include a wide range of topics, including the ever-evolving nature of AI technology, various ramifications for different industries, complicated ethical issues, resource limitations,

and user acceptance. Technology is developing at a previously unheard-of rate. The study's findings and conclusions could someday become obsolete if AI develops further. Our review generally concentrates on insights that are currently known to us in order to mitigate this constraint. We urge researchers and practitioners to keep up with the most recent advances in AI and to approach AI-infused design thinking from a futuristic standpoint. Different industries and design contexts may adopt and incorporate AI differently, with varying ramifications and difficulties. Although our study offers a comprehensive review of AI in the context of design thinking, it might not fully account for industry-specific peculiarities. Future studies should examine the application of AI in particular industries and its distinctive effects. The incorporation of AI raises complex, multidimensional ethical challenges. AI regulations and ethical issues are constantly changing. We relied on the most recent and trustworthy sources accessible as of our knowledge cutoff date to reduce potential bias. However, as AI technologies develop, staying alert and adapting to shifting ethical standards and laws are crucial. Particularly for small design teams or freelance designers, access to state-of-the-art AI technologies and resources can be constrained. This resource limitation may impact the degree to which AI may be practically incorporated into design processes. While noting this drawback, our study intends to draw attention to both the opportunities and the difficulties, highlighting the differences in resource access. Despite these drawbacks, this research provides a solid framework for further investigating the connection between AI and design thinking. Researchers, designers, and organizations interested in learning more about AI's direction in the design industry will find it a great resource. The drawbacks mentioned here highlight the necessity of continual investigation and modification as AI technology, ethical issues, and real-world applications of AI-infused design thinking continue to develop. This study promotes a flexible and futuristic approach to integrating AI into design thinking.

Ethical statement

Ethical approval is not applicable to this article.

Declaration of competing interest

The authors declare no conflicts of interest.

References

Abrams, M., Abrams, J., Cullen, P., Goldstein, L., 2019. Artificial intelligence, ethics, and enhanced data stewardship. IEEE Security & Privacy 17 (2), 17–30. https://doi.org/10.1109/MSEC.2018.2888778.

Achuthan, K., Nair, V.K., Kowalski, R., Ramanathan, S., Raman, R., 2022. Cyberbullying research—Alignment to sustainable development and impact of COVID-19: Bibliometrics and science mapping analysis. Comput. Hum. Behav. 140, 107566 https://doi.org/10.1016/j.chb.2022.107566.

Acklin, C., 2013. Design management absorption model: a framework to describe and measure the absorption process of design knowledge by SMEs with little or no prior design experience. Creativ. Innovat. Manag. 22 (2), 147–160. https://doi.org/10.1111/caim.12022.

Al Mesmari, S., 2023. Transforming data into actionable insights with cognitive computing and AI. J. Software Eng. Appl. 16 (6), 211–222. https://doi.org/10.4236/jsea.2023.166012.

Alghamdi, N.A., Al-Baity, H.H., 2022. Augmented analytics driven by AI: a digital transformation beyond business intelligence. Sensors 22 (20), 8071. https://doi.org/10.3390/s22208071.

Alshamrani, M., 2022. IoT and artificial intelligence implementations for remote healthcare monitoring systems: a survey. Journal of King Saud University-Computer and Information Sciences 34 (8), 4687–4701. https://doi.org/10.1016/j.jksuci.2021.06.005.

Ameen, N., Sharma, G.D., Tarba, S., Rao, A., Chopra, R., 2022. Toward advancing theory on creativity in marketing and artificial intelligence. Psychol. Market. 39 (9), 1802–1825. https://doi.org/10.1002/mar.21699.

As, I., Pal, S., Basu, P., 2018. Artificial intelligence in architecture: generating conceptual design via deep learning. Int. J. Architect. Comput. 16 (4), 306–327.

Badini, S., Regondi, S., Pugliese, R., 2023. Unleashing the power of artificial intelligence in materials design. Materials 16 (17), 5927.

Bleher, H., Braun, M., 2023. Reflections on putting AI ethics into practice: how three AI ethics approaches conceptualize theory and practice. Sci. Eng. Ethics 29 (3), 21.

Bresciani, S., 2015. The design process: a visual model. In: 2015 19th International Conference on Information Visualisation. IEEE, pp. 354–359.

Brock, J.K.U., von Wangenheim, F., 2019. Demystifying AI: what digital transformation leaders can teach you about realistic artificial intelligence. Calif. Manag. Rev. 61 (4), 110–134. https://doi.org/10.1177/1536504219865226.

Brown, N.C., Jusiega, V., Mueller, C.T., 2020. Implementing data-driven parametric building design with a flexible toolbox. Autom. ConStruct. 118, 103252 https://doi.org/10.1016/j.autcon.2020.103252.

Brown, T., 2008. Design thinking. Harv. Bus. Rev. 86 (6), 84.

Brownlee, J., 2019. Deep Learning for Computer Vision: Image Classification, Object Detection, and Face Recognition in python. Machine Learning Mastery.

Buchanan, R., 1992. Wicked problems in design thinking. Des. Issues 8 (2), 5–21. https://doi.org/10.2307/1511637.

Cao, J., Lam, K.Y., Lee, L.H., Liu, X., Hui, P., Su, X., 2023. Mobile augmented reality: user interfaces, frameworks, and intelligence. ACM Comput. Surv. 55 (9), 1–36.

Carlgren, L., 2013. Design thinking as an enabler of innovation: exploring the concept and its relation to building innovation capabilities. Chalmers Tekniska Hogskola (Sweden).

Carlgren, L., Rauth, I., Elmquist, M., 2016. Framing design thinking: the concept in idea and enactment. Creativ. Innovat. Manag. 25 (1), 38–57.

Carney, M., Webster, B., Alvarado, I., Phillips, K., Howell, N., Griffith, J., et al., 2020. Teachable machine: approachable Web-based tool for exploring machine learning classification. In: Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems, pp. 1–8. https://doi.org/10.1145/3334480.3382839.

Carvalho, A., Levitt, A., Levitt, S., Khaddam, E., Benamati, J., 2019. Off-the-shelf artificial intelligence technologies for sentiment and emotion analysis: a tutorial on using IBM natural language processing. Commun. Assoc. Inf. Syst. 44 (1), 43.

Chassang, G., Thomsen, M., Rumeau, P., Sèdes, F., Delfin, A., 2021. An interdisciplinary conceptual study of Artificial Intelligence (AI) for helping benefit-risk assessment practices. AI Communications 34 (2), 121–146.

Dang, T., Spathis, D., Ghosh, A., Mascolo, C., 2023. Human-centred artificial intelligence for mobile health sensing: challenges and opportunities. R. Soc. Open Sci. 10 (11), 230806.

de Spiegeleire, S., Maas, M., Sweijs, T., 2017. *Artificial intelligence and the Future of Defense: Strategic Implications for Small-And Medium-Sized Force Providers*—The Hague Centre for Strategic Studies.

de Villiers, R., 2022. Design thinking as a problem solving tool. In: The Handbook of Creativity & Innovation in Business: A Comprehensive Toolkit of Theory and Practice for Developing Creative Thinking Skills. Springer Nature Singapore, Singapore, pp. 223–242. https://doi.org/10.1007/978-981-19-2180-3_11.

Dennehy, D., Schmarzo, B., Sidaoui, M., 2022. Organising for AI-powered innovation through design: the case of Hitachi Vantara. Int. J. Technol. Manag. 88 (2–4), 312–334. https://doi.org/10.1504/IJTM.2022.121507.
Ebrahimi, A., 2023. Self-directed and self-designed learning: integrating imperative topics with case-COVID-19. Technologies.
Elsbach, K.D., Stigliani, I., 2018. Design thinking and organizational culture: a review and framework for future research. J. Manag. 44 (6), 2274–2306. https://doi.org/10.1177/0149206317744252.
Floridi, L., Cowls, J., Beltrametti, M., Chatila, R., Chazerand, P., Dignum, V., et al., 2021. An ethical framework for a good AI society: opportunities, risks, principles, and recommendations. Ethics, governance, and policies in artificial intelligence 19–39. https://doi.org/10.1007/978-3-030-81907-1_3.
Glen, R., Suciu, C., Baughn, C.C., Anson, R., 2015. Teaching design thinking in business schools. Int. J. Manag. Educ. 13 (2), 182–192. https://doi.org/10.5465/amle.2012.0308.
Haleem, A., Javaid, M., Qadri, M.A., Singh, R.P., Suman, R., 2022. Artificial intelligence (AI) applications for marketing: a literature-based study. International Journal of Intelligent Networks 3, 119–132. https://doi.org/10.1016/j.ijin.2022.08.005.
Hamington, M., 2019. Integrating care ethics and design thinking. J. Bus. Ethics 155 (1), 91–103. https://doi.org/10.1007/s10551-017-3522-6.
Han, J., Forbes, H., Shi, F., Hao, J., Schaefer, D., 2020. A data-driven approach for creative concept generation and evaluation. In: Proceedings of the Design Society: DESIGN Conference, vol. 1. Cambridge University Press, pp. 167–176. https://doi.org/10.1017/dsd.2020.5.
Harika, J., Baleeshwar, P., Navya, K., Shanmugasundaram, H., 2022. A review on artificial intelligence with deep human reasoning. In: 2022 International Conference on Applied Artificial Intelligence and Computing (ICAAIC). IEEE, pp. 81–84. https://doi.org/10.1109/ICAAIC53929.2022.9793310.
Harju, C., 2022. The perceived quality of wooden building materials—a systematic literature review and future research agenda. Int. J. Consum. Stud. 46 (1), 29–55. https://doi.org/10.1111/ijcs.12764.
Hassi, L., Laakso, M., 2011. Conceptions of design thinking in the management discourse. In: European Academy of Design Conference, pp. 341–351. Porto, Portugal, 4.-7.5. 2011.
Huang, M.H., Rust, R.T., 2021. Engaged to a robot? The role of AI in service. J. Serv. Res. 24 (1), 30–41.
Iandoli, L., 2023. Annual review article: the dual mindset of design-driven entrepreneurship: the case for a pedagogy of making and artefact-centred entrepreneurship education. Int. Small Bus. J. 41 (4), 349–370. https://doi.org/10.1177/02662426231160525.
Jarrahi, M.H., 2018. Artificial intelligence and the future of work: human-AI symbiosis in organizational decision making. Bus. Horiz. 61 (4), 577–586. https://doi.org/10.1016/j.bushor.2018.03.007.
Johansson-Sköldberg, U., Woodilla, J., Çetinkaya, M., 2013. Design thinking: past, present and possible futures. Creativ. Innovat. Manag. 22 (2), 121–146. https://doi.org/10.1111/caim.12023.
Johnson, J., Hurst, A., Safayeni, F., 2023. Managing data-driven design: a survey of the literature and future directions. Proceedings of the Design Society 3, 2525–2534. https://doi.org/10.1017/pds.2023.253.
Kelly, C.J., Karthikesalingam, A., Suleyman, M., Corrado, G., King, D., 2019. Key challenges for delivering clinical impact with artificial intelligence. BMC Med. 17, 1–9. https://doi.org/10.1186/s12916-019-1426-2.
Kirongo, A.C., Huka, G., Bundi, D., Kitaria, D., Muchiri, G., 2022. Implementation of AI-Based assistive technologies for learners with physical disabilities in areas of innovation and special schools: a practical design thinking approach. African Journal of Science, Technology and Social Sciences 1 (2), 73–76. https://doi.org/10.58506/ajstss.v1i2.124.
Knight, E., Daymond, J., Paroutis, S., 2020. Design-led strategy: how to bring design thinking into the art of strategic management. Calif. Manag. Rev. 62 (2), 30–52. https://doi.org/10.1177/0008125619897594.
Krishna, S.R., Rathor, K., Ranga, J., Soni, A., Srinivas, D., Kumar, A., 2023. Artificial intelligence integrated with big data analytics for enhanced marketing. In: 2023 International Conference on Inventive Computation Technologies (ICICT). IEEE, pp. 1073–1077. https://doi.org/10.1109/ICICT57646.2023.10134043.
Kumar, S., Sahoo, S., Lim, W.M., Dana, L.P., 2022. Religion as a social shaping force in entrepreneurship and business: insights from a technology-empowered systematic literature review. Technol. Forecast. Soc. Change 175, 121393. https://doi.org/10.1016/j.techfore.2021.121393.
Lee, D., Park, J.H., 2019. Future trends of AI-based intelligent systems and services: challenges, opportunities, and solutions. Journal of Information Processing Systems 15 (4), 717–723.
Liao, J., Hansen, P., Chai, C., 2020. A framework of artificial intelligence augmented design support. Hum. Comput. Interact. 35 (5–6), 511–544.
Liedtka, J., 2015. Perspective: linking design thinking with innovation outcomes through cognitive bias reduction. J. Prod. Innovat. Manag. 32 (6), 925–938. https://doi.org/10.1111/jpim.12163.
Liu, Q., Wang, K., Li, Y., Liu, Y., 2020. Data-driven concept network for inspiring designers' idea generation. J. Comput. Inf. Sci. Eng. 20 (3), 031004.
Lloyd, P., 2013. Embedded creativity: teaching design thinking via distance education. Int. J. Technol. Des. Educ. 23, 749–765. https://doi.org/10.1007/s10798-012-9214-8.
Magistretti, S., Ardito, L., Messeni Petruzzelli, A., 2021. Framing the microfoundations of design thinking as a dynamic capability for innovation: reconciling theory and practice. J. Prod. Innovat. Manag. 38 (6), 645–667. https://doi.org/10.1111/jpim.12586.
Mahato, S.S., Phi, G.T., Prats, L., 2021. Design thinking for social innovation: secrets to success for tourism social entrepreneurs. J. Hospit. Tourism Manag. 49, 396–406. https://doi.org/10.1016/j.jhtm.2021.10.010.
Mahmoud, M.M.S., 2022. A framework for leveraging artificial intelligence to improve industrial design practices. International Design Journal 12 (6), 257–269.
Marshall, S.P., 2009. Re-imagining specialized STEM academies: igniting and nurturing decidedly different minds, by design. Roeper Rev. 32 (1), 48–60. https://doi.org/10.1080/02783190903386884.
Micheli, P., Wilner, S.J., Bhatti, S.H., Mura, M., Beverland, M.B., 2019. Doing design thinking: conceptual review, synthesis, and research agenda. J. Prod. Innovat. Manag. 36 (2), 124–148. https://doi.org/10.3389/frai.2020.561802. Frontiers in Artificial Intelligence, 3, 561802.
Ouyang, C.H., Chen, C.C., Tee, Y.S., Lin, W.C., Kuo, L.W., Liao, C.A., et al., 2023. The application of design thinking in developing a deep learning algorithm for Hip fracture detection. Bioengineering 10 (6), 735. https://doi.org/10.3390/bioengineering10060735.
Paul, J., Lim, W.M., O'Cass, A., Hao, A.W., Bresciani, S., 2021. Scientific procedures and rationales for systematic literature reviews (SPAR-4-SLR). Int. J. Consum. Stud. 45 (4), O1–O16. https://doi.org/10.1111/ijcs.12695.
Quan, H., Li, S., Zeng, C., Wei, H., Hu, J., 2023. Big data and AI-driven product design: a survey. Appl. Sci. 13 (16), 9433.
Randhawa, K., Nikolova, N., Ahuja, S., Schweitzer, J., 2021. Design thinking implementation for innovation: an organization's journey to ambidexterity. J. Prod. Innovat. Manag. 38 (6), 668–700. https://doi.org/10.1111/jpim.12599.
Reay, S.D., Collier, G., Douglas, R., Hayes, N., Nakarada-Kordic, I., Nair, A., Kennedy-Good, J., 2017. Prototyping collaborative relationships between design and healthcare experts: mapping the patient journey. Design for Health 1 (1), 65–79.
Rezwana, J., Maher, M.L., 2022. Designing creative AI partners with COFI: a framework for modeling interaction in human-AI co-creative systems. ACM Trans. Comput. Hum. Interact. https://doi.org/10.1145/3519026.
Roberts, J.P., Fisher, T.R., Trowbridge, M.J., Bent, C., 2016. A design thinking framework for healthcare management and innovation. In: Healthcare, vol. 4. Elsevier, pp. 11–14. https://doi.org/10.1016/j.hjdsi.2015.12.002. No. 1.
Rosenschein, J.S., Zlotkin, G., 1994. Rules of Encounter: Designing Conventions for Automated Negotiation Among Computers. MIT press.
Sarker, I.H., 2021. Machine learning: algorithms, real-world applications and research directions. SN computer science 2 (3), 160. https://doi.org/10.1007/s42979-021-00592-x.
Sarker, I.H., 2022. Ai-based modeling: techniques, applications and research issues towards automation, intelligent and smart systems. SN Computer Science 3 (2), 158.
Schallmo, D., Williams, C.A., Lang, K., 2018. An integrated design thinking approach-literature review, basic principles and roadmap for design thinking. In: ISPIM Innovation Symposium. The International Society for Professional Innovation Management, pp. 1–18 (ISPIM).
Schneider, P., Walters, W.P., Plowright, A.T., Sieroka, N., Listgarten, J., Goodnow Jr, R.A., et al., 2020. Rethinking drug design in the artificial intelligence era. Nat. Rev. Drug Discov. 19 (5), 353–364. https://doi.org/10.1038/s41573-019-0050-3.

Seidel, V.P., Fixson, S.K., 2013. Adopting design thinking in novice multidisciplinary teams: the application and limits of design methods and reflexive practices. J. Prod. Innovat. Manag. 30, 19–33. https://doi.org/10.1111/jpim.12061.
Sharma, A.K., Singh, P., Vats, P., Jain, D., 2022. Deep learning and machine intelligence for operational management of strategic planning. In: Proceedings of Third International Conference on Computing, Communications, and Cyber-Security: IC4S 2021. Springer Nature Singapore, Singapore, pp. 475–485. https://doi.org/10.1007/978-981-19-1142-2_38.
Sharma, V., Panchal, A., Rane, V.Y., 2020. An analysis on current research trends and applications of natural language processing. Advance and Innovative Research 63.
Shi, Y., Gao, T., Jiao, X., Cao, N., 2023. Understanding design collaboration between designers and artificial intelligence: a systematic literature review. Proceedings of the ACM on Human-Computer Interaction 7 (CSCW2), 1–35.
Siahaan, D.T., Tan, C.S.L., 2022. What drives the adaptive capability of Indonesian SMEs during the Covid-19 pandemic: the interplay between perceived institutional environment, entrepreneurial orientation, and digital capability. Asian J. Bus. Res. 12 (2).
Simon, H.A., 1995. Artificial intelligence: an empirical science. Artif. Intell. 77 (1), 95–127.
Skov, S.S., Andersen, J.R., Lauridsen, S., Bab, M., Bundsbæk, M., Nielsen, M.B.D., 2022. Designing a conversational agent to promote teamwork and collaborative practices using design thinking: an explorative study on user experiences. Front. Psychol. 13, 903715 https://doi.org/10.3389/fpsyg.2022.903715.
Solakis, K., Katsoni, V., Mahmoud, A.B., Grigoriou, N., 2022. Factors affecting value co-creation through artificial intelligence in tourism: a general literature review. J. Tourism Futur. https://doi.org/10.1108/JTF-06-2021-0157.
Spring, M., Faulconbridge, J., Sarwar, A., 2022. How information technology automates and augments processes: insights from Artificial-Intelligence-based systems in professional service operations. J. Oper. Manag. 68 (6–7), 592–618. https://doi.org/10.1002/joom.1215.
Sreenivasan, A., Suresh, M., 2022. Future of healthcare start-ups in the era of digitalization: bibliometric analysis. International Journal of Industrial Engineering and Operations Management. https://doi.org/10.1108/IJIEOM-10-2022-0046 (ahead-of-print).
Stickdorn, M., Hormess, M.E., Lawrence, A., Schneider, J., 2018. This Is Service Design Doing: Applying Service Design Thinking in the Real World. O'Reilly Media, Inc.
Stiglic, G., Kocbek, P., Fijacko, N., Zitnik, M., Verbert, K., Cilar, L., 2020. Interpretability of machine learning-based prediction models in healthcare. Wiley Interdisciplinary Reviews: Data Min. Knowl. Discov. 10 (5), e1379 https://doi.org/10.1002/widm.1379.
Thakur, P., Dhar, R.L., 2022. Impact of diversity training on employees and consumers: a review and research agenda. Int. J. Consum. Stud. 46 (5), 1665–1690. https://doi.org/10.1111/ijcs.12823.
Trocin, C., Stige, Å., Mikalef, P., 2023. Machine Learning (ML) diffusion in the design process: a study of Norwegian design consultancies. Technol. Forecast. Soc. Change 194, 122724. https://doi.org/10.1016/j.techfore.2023.122724.
Uebernickel, F., Jiang, L., Brenner, W., Pukall, B., Naef, T., Schindlholzer, B., 2020. Design Thinking: the Handbook. World Scientific.
Verganti, R., Vendraminelli, L., Iansiti, M., 2020. Innovation and design in the age of artificial intelligence. J. Prod. Innovat. Manag. 37 (3), 212–227. https://doi.org/10.1111/jpim.12523.
Vilar, E., Filgueiras, E., Rebelo, F. (Eds.), 2022. Virtual and Augmented Reality for Architecture and Design. CRC Press.
Visser, M., van Eck, N.J., Waltman, L., 2021. Large-scale comparison of bibliographic data sources: Scopus, Web of science, Dimensions, crossref, and microsoft academic. Quantitative Science Studies 2 (1), 20–41. https://doi.org/10.1162/qss_a_00112.
Wamba-Taguimdje, S.L., Fosso Wamba, S., Kala Kamdjoug, J.R., Tchatchouang Wanko, C.E., 2020. Influence of artificial intelligence (AI) on firm performance: the business value of AI-based transformation projects. Bus. Process Manag. J. 26 (7), 1893–1924. https://doi.org/10.1108/BPMJ-10-2019-0411.
Wang, Y., Chiew, V., 2010. On the cognitive process of human problem solving. Cognit. Syst. Res. 11 (1), 81–92. https://doi.org/10.1016/j.cogsys.2008.08.003.
Wolferts, D., Stein, E., Bernards, A.K., Reiners, R., 2022. Differences between remote and analog design thinking through the lens of distributed cognition. Frontiers in Artificial Intelligence 5, 915922. https://doi.org/10.3389/frai.2022.915922.
Worsley, M., Bar-El, D., 2022. Inclusive Making: designing tools and experiences to promote accessibility and redefine making. Comput. Sci. Educ. 32 (2), 155–187.
Xu, B., Jiang, J., 2022. Exploitation for multimedia asian information processing and artificial intelligence-based art design and teaching in colleges. ACM Transactions on Asian and Low-Resource Language Information Processing 21 (6), 1–18. https://doi.org/10.1145/3526219.
Yeung, K., Howes, A., Pogrebna, G., 2020. AI Governance by Human Rights–Centered Design, Deliberation, and Oversight, AI. The Oxford handbook of ethics of, pp. 77–106.
Yuan, T., Wang, Z., Rau, P.L.P., 2023. Design of intelligent real-time feedback system in online classroom. In: International Conference on Human-Computer Interaction. Springer Nature Switzerland, Cham, pp. 326–335. https://doi.org/10.1007/978-3-031-35946-0_27.
Zhou, C., Zhang, X., Yu, C., 2023. How does AI promote design iteration? The optimal time to integrate AI into the design process. J. Eng. Des. 1–28.
Zhu, Q., Luo, J., 2023. Generative transformers for design concept generation. J. Comput. Inf. Sci. Eng. 23 (4), 041003 https://doi.org/10.1115/1.4056220.